

Two Maps, Then None: Inside Alberta's 2026 Boundary Audit

A plain-language summary of the 2025–26 Electoral Boundary Commission forensic audit, and what the audit will measure next.
v0.22 — April 25, 2026 — [*Full monograph*](#)

An Airdrie resident pulls up the new electoral map on her phone. Her street goes to one riding. Two blocks over goes to another. Under the minority commission proposal, her city of 84,000 is cut into four pieces, each quarter attached to a different surrounding region. Nothing about Airdrie changed. Only the map did.

This audit is about that map, a second one the same commission produced, and a third map a legislative committee in Edmonton is drawing right now.

The short version

Alberta's Electoral Boundary Commission finished its work on March 23, 2026 and could not agree. Three commissioners produced one map; the other two produced a different one. Both are legal under the *Electoral Boundaries Commission Act*. This audit measured both using the same methods, applied identically. Three findings stand out.

1. **The two maps differ on six things you can measure without looking at any election results:** how evenly people are spread across districts, whether voters are concentrated, how badly cities are cut up, whether borders follow city limits, the shape of the districts, and how many trouble spots the commission's own chair flagged. The minority map differs from the majority on every one of them.
2. **The direction runs against the places where the governing party is weakest.** Calgary's northwest quadrant, the City of Airdrie, urban areas with established municipal boundaries — the communities most affected by the minority map's structural differences are the same ones where the governing party's opponents are strongest. The audit cannot determine intent. It can measure effect.
3. **The process now promoting the minority map has no precedent in Canada.** No other province lets a cabinet hand redistricting to a committee its own party controls partway through a redistribution

cycle. Most provinces either require the legislature to debate the commissioners' map first, or give the commission's map automatic effect unless overridden. Alberta does neither. On April 16, the government set both commission maps aside and assigned the work to a five-MLA committee, three from the governing UCP.

The process is its own finding, separate from the maps.

The April 16 decision

The *Electoral Boundaries Commission Act* lets the government refer the commission's recommendations to a legislative committee. On April 16, the government did this. The new committee — chaired by Brandon Lundy, UCP MLA for Leduc-Beaumont — has five members. Three from the governing UCP, one from the NDP, one from the Alberta Party. It must report by November 2, 2026 with a 91-seat map.

The referral followed the letter of the Act. It also confirms Alberta's outlier status: most provinces require the legislative assembly to consider a commission report before any referral can take effect; Alberta does not. Most provincial commissions are chaired by a sitting or retired judge whose tenure is protected; Alberta's chair is a public appointee who can be replaced mid-mandate by a cabinet decision.

Three Canadian comparator pivots show what "departing from norm" looks like in practice:

- **Quebec, 1992:** the National Assembly debated and amended the commission's report, then passed it. The drafting process stayed with the commissioners; the legislature only changed specific lines, on the record.
- **Ontario, 1996:** Mike Harris's government cut 130 ridings to 103 by adopting the federal map. A structural change, not a partisan one, with cross-party committee participation.
- **British Columbia, 2008:** a tied commission tabled both reports; MLAs picked one on a free vote. The drafting was not handed to a new body.

None of the three substituted a government-chaired drafting body for the commission partway through the cycle. That is what Alberta did on April 16.

Three things the minority map does differently

It splits the City of Airdrie into four pieces. The law caps each electoral division at one-and-a-quarter times the provincial average, so Airdrie needs at least two divisions. The majority map gives it two. The minority gives it four — North, East, West, and centre — each one stapled to a different rural or Calgary-edge district. An Airdrie resident with a question for her MLA has to know which quarter of the city she lives in before she can call the right office. Her neighbours two blocks over will give her three different answers.

SIDEBAR — WHY AIRDRIE MATTERS

Population 84,000. Larger than Red Deer. One council, one tax bill, one school division.

Majority map: 2 divisions.
Minority map: 4 — each quarter attached to a different surrounding region.

Both maps are legal. The four-way split is a choice.

It ignores municipal boundaries. When electoral maps follow the edge of a city or town, voters recognize where their division begins and ends — the property-tax line, the school-division line, the local-election ward line, and the provincial-election line all coincide. These maps already exist. Statistics Canada publishes the city and town boundaries; any redistricting commission gets them for free. The majority map traces them on 71 of every 100 kilometres of its border. The minority traces them on 15 of 100 — and offers no published reason for the departure.

One area of Calgary is carved up to concentrate NDP voters into larger-than-average divisions. In Calgary's northwest quadrant, the minority map's divisions average 11.5% above the province-wide population — versus 2.8% on the majority. The same geographic zone, drawn by the same commission under the same constraints, produces districts a quarter larger on one map than on the other. Packing more voters into fewer, larger districts is one of the two classic gerrymandering moves: it dilutes how much each ballot weighs.

Two further structural measures show the same pattern: the minority map's overall population spread is 48% wider than the majority's (median absolute deviation 4,707 vs 3,180), and the commission chair flagged three boundaries on the minority map as geographically anomalous (Rocky Mountain House–Banff Park's extension into uninhabited national park land; the Calgary–Nolan Hill–Cochrane lasso-shaped corridor; the Olds–Three Hills–Didsbury reach into north Airdrie). The majority received zero such flags from the same chair.

What the audit found across five measures

The table compares the two maps. None of the top six rows uses election results.

SIDEBAR — VOCABULARY
PACKING / CRACKING.

Packing concentrates one party's voters into a few high-margin wins (wasting their surplus). Cracking spreads them thinly across districts they narrowly lose. Both shrink the party's seat count below its vote share.

ANCHORING.

The fraction of an electoral border that lies on a pre-existing administrative line — a city limit, a school-division boundary, a Statistics Canada census line. The reference geography is published; commissions get it for free.

COMPUTER SIMULATION.

The audit generated 250,000 different legal Alberta maps under the same statutory rules the commission worked under, then compared the real maps to the simulated distribution. A real map at the 95th percentile or beyond sits in the outermost 5% of what those constraints can produce.

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
How evenly distributed the populations are	More even	48% more uneven
NW Calgary population excess above average	2.8%	11.5%
Airdrie split	2 divisions	4 divisions
Borders that follow existing municipal lines	71%	15%
Borders anchored to Statistics Canada census boundaries	80%	17%
Boundaries flagged by the commission chair	0	3
Efficiency gap (a partisan-fairness measure)	-1.3%	-2.7%
Packing-cracking neighbourhood pattern	3 (same as 2019)	0 — <i>pre-registered pass</i>

The bottom two rows depend on election results and are less certain. Both efficiency-gap numbers favour the UCP relative to a perfectly neutral outcome — which is expected in Alberta, where concentrated urban NDP support produces a structural conservative tilt under any map. Both numbers fall well below any published gerrymander threshold, including the four Alberta-calibrated thresholds the audit derived (see methodology).

The last row is where the minority map looks *better*: zero packed-cracked adjacency pairs, against three on the majority and three on the current 2019 map — a pre-registered pass.

How the audit measures gerrymandering

A gerrymander is a deliberate boundary choice that advantages one political interest over others, beyond what the statutory constraints require. Two methodological problems make this hard to measure honestly.

The first is that the redistricting problem is mathematically *NP-hard*: there is no single correct Alberta map. Any map satisfying the *Electoral Boundaries Commission Act* — population deviation within $\pm 25\%$, contiguity, compactness, community-of-interest preservation, Indigenous effective representation, and public-hearing input — is one of an astronomical family of legal maps. Fairness in this domain is not a discovery; it is a choice among many legitimate alternatives.

The second is that intent cannot be observed from a map. A commissioner could draw a partisan-skewed map either deliberately or by unlucky chance among the legal alternatives. The audit cannot read minds. What it can do is measure *effects* — how the map a commission produced compares to the family of maps a non-partisan procedure would have drawn.

The audit handles both problems with three layered tools.

A 100,000-map computer simulation. Using a redistricting algorithm called *ReCom* (Recombination Markov Chain Monte Carlo), the audit generates 100,000 different legal Alberta maps from the same statutory constraints the commission worked under. Each simulated map is scored on the four standard partisan-bias metrics. The real maps' scores are then placed into this distribution: a real map at the 5th or 95th percentile is a statistical outlier; a map between the 25th and 75th percentile sits in the typical range. The minority sits at the 95th-percentile boundary on mean-median; the majority sits at the same boundary. The 2019 enacted map sits at the same boundary. This tells the audit what is geography (all three baselines tilt the same way) versus what is drawing (the minority's structural irregularities, where the majority and 2019 do not show the same pattern).

Four Alberta-calibrated thresholds for the efficiency-gap test. The famous 7% threshold from US case law was calibrated to US Congressional delegation sizes and never adopted by any court. The audit derives four Alberta-specific alternatives: a first-principles assembly-size sensitivity threshold ($\sim 2.2\%$), an EBCA statutory-proportional threshold (5%), an Alberta historical-swing threshold (5–9% provisional), and the strongest — the MCMC ensemble's 95th-percentile EG of 4.37%, drawn from Alberta's actual 2019 substrate under EBCA constraints. Under partial-coverage geometry (67 of 89 EDs measurable), both 2026 maps fell below all four thresholds. Under the audit's now-current full-coverage geometry (89 of 89, via 2019-Tier-A inheritance fill for the 33 EDs whose 2026 boundaries could not be directly reconstructed), both maps exceed the 4.37% threshold (majority 6.4%, minority 9.2%) and both score in the top 0.1% of a 250,000-plan computer simulation. Multi-method disagreement on this magnitude is itself a finding — see Lane 1 below.

A pre-registered structural-irregularity scorecard. Five tests, committed to in writing before any results were read: anchoring departure from Canadian comparator norm, population MAD widening, geographic-zone gap, chair-flagged anomaly count, and rationale-failure count under symmetric audit. A map crossing four of five qualifies as a structural-irregularity outlier under the audit's own pre-registered rule. The minority crosses five of five. The majority crosses zero of five.

The combination of these three tools is what produces the two-lane verdict in the next section. No single tool is dispositive. Directional consistency across all three is the audit's inferential standard.

One way to read the scorecard

Rows of measurements are hard to weigh against each other. One defensible synthesis is a four-category rubric weighted by the principles redistricting law and democratic theory treat as primary.

SCORECARD — ONE WAY TO READ THE TWO MAPS

A proposed synthesis with explicit weights — not a finding. Other weightings give different grades.

CATEGORY	WEIGHT	MAJORITY	MINORITY
Voter equality	25%	B	D
Community recognition	35%	B	F
Map coherence	25%	A	D
Partisan fairness	15%	B	B
OVERALL	—	B (87/100)	F (47/100)

Why these weights. Community recognition carries the most (35%) because *Reference re Saskatchewan* — the Supreme Court decision defining what redistricting is for — holds that effective representation requires districts that communities can identify as their own. The Act's $\pm 25\%$ population band grounds voter equality at 25%. Map coherence (25%) reflects the commission's own standard: the chair flagged specific boundaries as anomalous, and those flags matter. Partisan fairness is weighted lowest (15%) because both maps fall below any published gerrymander threshold, and the direction of the partisan-fairness finding reverses depending on which measurement method you use.

What produces the minority's F. On community recognition (10/35 under these weights), 14.5% municipal anchoring means 85 out of every 100 kilometres of the minority map's border cuts through territory voters do not recognize from any other civic map — not their school division boundary, not their property-tax bill, not their local election ward. The majority's equivalent is 71%. The Airdrie four-way split compounds the effect — residents must navigate four legislative offices, each primarily identified with somewhere else. Three chair-flagged anomalies — a lasso-shaped corridor, a boundary extension through uninhabited national park land, and a district named for communities together smaller than the community it actually captures — pull the map-coherence score down further.

Where the maps score comparably. Under the lowest-weighted category — partisan fairness — the two maps are effectively tied, and neither shows a serious problem by any published standard.

Was either map a gerrymander?

The honest answer needs two lanes.

LANE 1 — THE PARTISAN-FAIRNESS NUMBERS

The 4.37% number we keep referring to is not a "threshold" the way *Whitford v. Gill* called 7% a threshold. It is the 95th percentile of the efficiency gap across 250,000 different legal Alberta maps the audit's computer simulation generated. Anything above it sits in the outermost 5% of what Alberta's redistricting rules can produce. That is informative, but it is a *percentile*, not a legal cut-off.

Where the two 2026 maps land against the simulation's 95th percentile (4.37%):

 Lane 1: efficiency gap before/after the coverage shift. Faded dots show the partial-coverage reading (Majority -1.29%, Minority -2.71%); bold dots show the full-coverage reading (Majority +6.43%, Minority +9.21%). Arrows mark the partial→full shift. Both bold dots sit beyond the dashed p95 = 4.37% line.

MAP	EFFICIENCY GAP	ABOVE THE P95 LINE?	HONEST PERCENTILE READING
Majority 2026	+6.4%	Yes — 2.0 pp over	beyond p95 (likely p98+)
Minority 2026	+9.2%	Yes — 4.8 pp over	beyond p95 (likely p99+)

Why "likely" instead of an exact number. The chains' effective sample size is in the high hundreds, which means the audit can confidently say a map is past p95 but cannot reliably distinguish p99 from p99.9 from p100. The point estimate is in the data; the precision the audit can claim stops at "outside the 95–5 band."

Why the verdict moved. An earlier version of this audit measured 67 of 89 ridings — the 22 it skipped were mostly rural UCP strongholds whose 2026 boundaries the commission's published maps did not let the audit reconstruct from public data. Filling those in (using their 2019 boundaries as a stand-in until Elections Alberta releases the official files) pushed both maps over the line. The earlier reading wasn't wrong — it was incomplete. The honest framing: the partial-coverage and the inheritance-filled full-coverage readings *both* have measurement noise; until the official shapefiles drop, neither is authoritative. They agree on direction (both maps lean UCP) and disagree on magnitude.

Lane 2 — the structural pattern

Lane 1 depends on which election and which version of the audit's geometry you use. Lane 2 does not. The structural evidence is in the maps themselves — drawn lines, split cities, where the boundaries do and don't follow administrative lines that exist for other reasons. On these tests, the two maps are not close.

 Lane 2: structural-irregularity tests, majority (blue) vs minority (orange). Each row's dashed red mark shows that test's threshold; the minority bars (orange) extend visibly past every threshold; the majority bars (blue) are flush with zero or well within the norm.

TEST	MAJORITY MAP	MINORITY MAP
Border follows existing municipal lines (70–85% Canadian norm)	71% — within norm	15% — 4.9× below norm
Population spread (tighter is better)	3,180	4,707 — 48% wider
NW Calgary population concentration above average	0.4%	12.2%
Boundaries flagged by the commission's own chair	0	3
Airdrie split (constraint minimum: 2)	2 pieces	4 pieces
Pre-registered structural-irregularity count (4 of 5 = outlier)	0 of 5	5 of 5

TEST	MAJORITY MAP	MINORITY MAP
Mean-median percentile vs computer simulation	At the 95th-percentile boundary	At the 95th-percentile boundary

A separate finding – applied only to the minority because the minority is the map whose contested redraws are public – is that **six of seven of the minority commissioners' published reasons fail under independent check**, and **two of the contested redraws have no documented public-submission support** in the commission's archive of 1,140+ submissions. These rows are not in the table because they cannot be applied symmetrically: the majority did not publish a contested-redraw rationale list. They are reported here as a single flag, not as two more rows in the structural-irregularity count.

On Lane 2, the majority crosses zero structural thresholds. The minority crosses every one of them by a wide margin.

Six of seven of the minority's published reasons fail under check

The minority report includes published rationales for its seven most contested redraws. The audit checks each one against the data the rationale itself invokes. Six fail; one stands.

- **Airdrie split four ways.** *Minority commissioners said:* rural and urban character differs across Airdrie's quadrants. *Statistics Canada says:* the four quadrants are within 8% of each other on every demographic measure the commission considers. **Fail.**
- **Cochrane attached to Calgary-Foothills via the Nolan Hill lasso.** *Minority commissioners said:* commuter community of interest. *Journey-to-work data says:* Cochrane's most common commuter destination is Cochrane itself; among out-of-town commuters, Calgary-Centre is named more often than Foothills. **Fail.**
- **Rocky Mountain House extended into Banff National Park.** *Minority commissioners said:* ranching community of interest. *The extended polygon contains:* zero year-round residents and zero working ranches. **Fail.**
- **Olds-Three Hills-Didsbury reaching into North Airdrie.** *Minority commissioners said:* linked agricultural service centres. *Census-of-agriculture data says:* North Airdrie is suburban; Olds and Didsbury are rural service centres. The connection is geographic adjacency, not common interest. **Fail.**
- **Lethbridge split between Lethbridge-East and Taber-Warner.** *Minority commissioners said:* matches federal boundary. *Elections Canada records say:* the federal boundary they claim to match was retired in 2013. **Fail.**
- **Red Deer attached to Sylvan Lake.** *Minority commissioners said:* shared school division. *The shared school division contains:* 8% of Red Deer's school-aged population. **Fail (on magnitude).**
- **St. Albert-Sturgeon.** *Minority commissioners said:* population balancing on Edmonton-corridor constraints. *The constraint check confirms:* with the population-deviation rule and the Edmonton-corridor commute geography, no other configuration satisfies both the community-of-interest and the $\pm 25\%$ rule simultaneously. **Stands.**

The six-failure / one-stand pattern matters. A symmetric audit finding 0 of 7 or 7 of 7 would suggest either no rationale problem or a methodology bias. A 6 of 7 pattern with the seventh being constraint-forced is the engineered-versus-forced ratio that distinguishes deliberate boundary choice from a trade-off the constraints made unavoidable.

Verdict

 Two-by-two verdict quadrant. Lane 1 (efficiency gap) on the x-axis; Lane 2 (structural-irregularity count out of 5) on the y-axis. The 2019 baseline sits in the bottom-left (clean on both lanes). The Majority 2026 dot sits beyond the EG p95 line but at zero on the structural axis. The Minority 2026 dot sits in the top-right corner — beyond threshold on both lanes.

	LANE 1: NUMBERS (FULL COVERAGE)	LANE 2: STRUCTURE
Majority 2026	EG +6.4% (beyond p95)	clean — crosses no structural threshold
Minority 2026	EG +9.2% (beyond p95)	crosses every structural threshold by a wide margin

The Lane 1 reading moved when the audit went from 67-of-89 to 89-of-89 coverage; under the earlier partial reading, neither map was beyond p95. Both readings agree on direction (both maps lean UCP) and on which map leans further (the minority). The engineering difference that distinguishes the two maps lives in Lane 2, not Lane 1 — both maps inherit Alberta's structural rural-UCP advantage on partisan-fairness numbers; only the minority crosses every structural threshold.

THE PLAIN READING.

On the numbers, both maps now look like outliers — once you measure all 89 ridings instead of 67. On the structural pattern, only the minority map crosses every line by a wide margin. Each of those structural choices has an innocent explanation on its own; together, they form the kind of pattern redistricting researchers flag for further inquiry. Whether the cause was deliberate engineering, unlucky drafting, or both is a judgement call the audit puts in the reader's hands.

We measured the effects. We can't read minds.

How close did the minority map come?

- **On the partisan-fairness numbers (full coverage):** about 4.8 percentage points OVER the Alberta-calibrated 4.37% threshold (minority EG +9.2%). The majority is also over by 2.0 pp (+6.4%). Both maps land in the top 0.1% of a 250,000-plan computer simulation. Note that the partial-coverage reading put both maps under threshold; the full-coverage shift is itself a finding.
- **On the structural pattern:** already over every threshold the audit applies, by wide margins. The 4.9× anchoring departure from Canadian norm is the single largest threshold cross the audit measured.

The minority map maxes out the structural-irregularity scoring while staying inside the partisan-fairness numbers band.

SCORECARD — RETRACTION CONDITIONS

If any of these turn out to be true, the verdict above is wrong and gets retracted publicly within 30 days.

1. A COUNTER-MAP EXISTS.

Someone produces a legal Alberta map satisfying the minority's own community-of-interest reasons (Airdrie, Cochrane, Nolan Hill, Rocky Mountain House–Banff Park) *and* anchoring on municipal boundaries at majority-comparable rates. Open challenge — Issue #14 on the audit's GitHub repository.

2. THE LUNTY COMMITTEE MAP LOOKS LIKE THE MINORITY.

The November 91-seat map produces structural-irregularity counts in the same range — suggesting Alberta-specific drawing difficulty rather than minority-specific engineering.

3. A PRE-2026 INTERNAL COMMISSION DOCUMENT SURFACES.

Showing the minority's choices were a deliberate response to documented community submissions rather than drafting choices.

4. THE CROSS-ELECTION REVERSAL EXTENDS TO RECENT POLLING.

The structural pattern would become harder to read as engineering if the partisan direction is genuinely vote-distribution-dependent.

What this audit is now for

In policy terms, the verdicts above are moot. On April 16 the government set both commission maps aside; Alberta's 2027 map will come from the Lundy committee. Neither commission map is a candidate to become law.

The work itself is not moot. The commission audit calibrated Alberta-specific thresholds — structural-irregularity tests, anchoring comparators, a 100,000-plan computer ensemble — against three real Alberta maps (the 2019 enacted boundaries plus the two 2026 commission proposals, applied symmetrically). That framework will be re-pointed at the Lundy committee's map the day it is published, reported by the same standard you have just read.

The standard for November is the standard applied here: *skew toward neutrality wherever the constraints permit, document the departures, and defend them with evidence stronger than aesthetic preference*. When the Lundy committee publishes its map in November, this audit will measure it the same way, with the same code, against the same thresholds. The results will be public the day the map is.

**SIDEBAR — TWO
REMINDERS FOR THE
NOVEMBER AUDIT**

Alberta's geography will always favour conservatives. NDP voters concentrate in city cores; UCP voters spread across efficient-margin rural ridings. A neutrally-drawn Alberta map produces a UCP-favourable efficiency gap as a starting point — that's geography, not gerrymandering.

Many things count as gerrymandering, and not all show up in the partisan-fairness numbers. The minority map sat sub-threshold on every Alberta-calibrated EG line while crossing every structural threshold by a wide margin. A drafting process that wants to engineer outcomes without leaving an EG fingerprint has the structural lane available — community splits, off-reference borders, anchoring departures, chair-flagged shapes.

What the audit does not claim

The audit does not claim the minority map was drawn to help the UCP win seats — intent cannot be proven from maps alone. It claims only that the differences are measurable, consistent, and not explained by Alberta's geography.

Both maps satisfy the *Electoral Boundaries Commission Act*.

The partisan-fairness numbers are sensitive to which election's results you use. Two different measurement methods give answers that point in opposite directions. The audit reports both, explains why they disagree, and does not pick one as correct. This is not evasion — it is an honest statement of the limits of what public data can tell you without the official map files, which Elections Alberta has not released.

The audit does not reach a constitutional conclusion. The *Reference re Saskatchewan* [1991] effective-representation framework is the standard a court would apply; whether the evidence here supports a challenge under that standard is for counsel and the courts to assess.

Behind the audit

This audit was produced by Will Conner, a fourth-year computer information systems student at

Mount Royal University in Calgary. It is an independent research project with no funding from any political party or advocacy organization. The author is a UCP-disinclined Alberta voter; that prior is disclosed in the monograph's Author Disclosure block, along with the three findings retained in the paper that ran *against* the prior.

The audit used publicly available data: Elections Alberta's 2023 Statement of Vote (vote totals by voting area), Statistics Canada's 2021 Census (population, dissemination areas, census subdivisions), and the commission's 2026 final report (per-ED population tables, published rationales, chair-flag annotations). All code is published in the GitHub repository and can be run independently by any reader. The pre-registered checklist that drove the audit was published before any 2026 results were measured; it is timestamped via Open Science Framework with embargoed release scheduled for 2026-11-02 (the day the Lundy committee must report).

The Elections Alberta shapefiles for the 2026 proposed boundaries have not been released. The audit reconstructs them from the commission's 600-DPI map images using a derived-provisional-geometry pipeline (eight stages, documented in monograph §4.1.5–§4.1.6). The reconstruction has documented $\pm 500\text{m}$ perimeter precision and Tier-dependent area precision; every metric in this article that depends on geometry carries that caveat in the full monograph. A sunset clause binds the audit to recompute all geometry-dependent findings within 48 hours of any official shapefile release.

AI assistance disclosed: Claude (Anthropic) helped draft and check the text. All factual claims were verified by the author against primary sources.

What you can do

1. **Read the full findings.** The monograph ([report_academic.md](#)) contains every number, every source, and every caveat. The retraction conditions tell you exactly what evidence would change each conclusion.
2. **Ask your MLA three questions about the committee.** – Will it publish the criteria it's using *before* it draws a single line? – Will it release the other maps it considered, not just the one it picked? – Will it disclose any AI tools it used and the prompts given to them?
3. **Request the official shapefiles.** Elections Alberta has not published the 2026 electoral division map files. Without these, some of this audit's calculations cannot be fully verified or corrected. A Freedom of Information request would require their release.

The full technical monograph, with all methods, caveats, and source citations, is at [report_academic.md](#).